



SSAP Enterprise Platform

Deterministic AI Governance Infrastructure

Cost optimization is a consequence. Governance is the foundation.

SSAP is a governance-first execution layer for LLM systems. It brings deterministic control, claim-level verification, and full audit replay to enterprise AI deployments.

1. The Real Enterprise Problem

Large Language Models are probabilistic systems. Enterprise environments are not.

As AI adoption scales, organizations face structural risks:

- Non-reproducible model behavior
- Confident but unverified factual claims
- Limited visibility into decision paths
- Regulatory exposure (EU AI Act, GDPR, internal policy frameworks)
- Vendor dependency without governance control

AI without a control layer becomes operational liability.

2. The SSAP Governance Model

SSAP introduces a deterministic execution layer between your applications and any LLM provider. It consists of two tightly integrated components:

A. SSAP Core — Decision Control

- Formal execution graph (PROBE → FULL → BLOCK)
- Bounded escalation (max 1 retry)
- Policy-versioned routing
- Vendor-agnostic model integration
- Structured inference paths

SSAP controls when and how inference is allowed to occur.

B. Integrity Engine — Claim-Level Enforcement

Integrity verifies output at the claim level before release. In Required Mode:

- No factual claim leaves the system without evidence reference
- Each claim has verification status (PASS / FAIL)
- FAIL blocks output
- All claims are logged for audit

Integrity controls what is allowed to leave the system.

C. Deterministic Audit & Replay

Every request produces:

- decision_path
- policy_version
- evidence snapshot reference
- integrity_result

This enables:

- Full replay under identical conditions
- Policy traceability
- Compliance-ready audit logs
- Drift detection and monitoring

SSAP controls why a decision was made.

3. Governance Execution Model

```
INPUT
|
v
SSAP_DECISION (policy-versioned)
|-- PROBE --> LLM_PROBE --> INTEGRITY_CHECK
|               |-- PASS --> RETURN
|               +-- FAIL --> ESCALATE
|
+-- FULL --> LLM_FULL --> INTEGRITY_CHECK
|               |-- PASS --> RETURN
|               +-- FAIL --> BLOCK
```

Properties:

- No uncontrolled inference loops
- No hidden retries
- Deterministic escalation boundaries
- Replayable execution



4. Enterprise Guarantees

SSAP guarantees:

- ✓ No unverified factual claims in Required mode
- ✓ No uncontrolled runtime inference paths
- ✓ Versioned and reproducible decision logic
- ✓ Complete audit trace of AI decisions
- ✓ Vendor-agnostic governance control

SSAP does not claim:

- ✗ Zero model errors
- ✗ Replacement of professional judgment
- ✗ Guaranteed external business outcomes

5. Beyond Cost Optimization

While SSAP typically reduces LLM spend by 60–70%, cost efficiency is a structural outcome of governance:

- **NO_INFERENCE** paths eliminate unnecessary calls
- **PROBE** paths reduce overuse of full inference
- Risk-aware escalation prevents waste
- Structured routing improves model utilization

Optimization emerges from control — not vice versa.

6. Enterprise Capabilities

Security & Compliance	Operations	Integration
• EU AI Act support • Zero raw prompt storage • TLS 1.3 / AES-256 • RBAC & tenant isolation • Complete audit trail	• Real-time dashboard • Quality monitoring • Drift detection • Policy engine • Kill-switch capability	• Drop-in proxy • OpenAI-compatible • Multi-provider support • Streaming support • < 15ms overhead

7. What SSAP Is

SSAP is not a chatbot layer. It is not a cost proxy. It is not a moderation filter.

SSAP is infrastructure for governing AI systems in production.

It introduces deterministic execution control over probabilistic intelligence.

Contact	Resources
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